

Student Enrollment & Course Management System (SECMS)

Project Documentation

A Salesforce-Based Academic Enrollment Management Application

Submitted as a Group Project under the TN Skills - Naan Mudalvan Programme

Project Overview and Scenarios

Group ID: SWTID-2026-4095 | Islamiah Women's Arts and Science College | Thiruvalluvar University

Project Description

SECMS is a Salesforce-based academic management solution that brings student, course, instructor and enrollment information into one cloud environment. It uses declarative automation, validation rules, an approval process, reports, dashboards and security controls to support the enrollment lifecycle.

Objective

Replace manual and fragmented enrollment handling with a structured, scalable, automated and secure system that improves data consistency and makes enrollment status easier to track.

Scope

Data objects, fields, relationships, Lightning navigation, record types, page layouts, validation rules, approval processing, flows, reports, dashboards and security controls.

Key Scenarios

- Admin creates a student record with a valid email.
- Admin maintains courses and instructors.
- An enrollment is created and routed through automation and approval.
- Admin reviews reports and the dashboard for operational status.

Technical Architecture and Workflow

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Architecture

- Presentation: Salesforce Lightning App, tabs and record pages.
- Application / Automation: Flows, Approval Process and Validation Rules.
- CoreData: Student, Course, Instructor and Enrollment custom objects.
- Analytics: Reports and Student Management Dashboard.
- Security: Profiles, Permission Sets and Sharing / OWD.

Relationships

Enrollment is the central transaction object. It references Student, Course and Instructor through lookup relationships so the enrollment lifecycle can be viewed in context.

Project Workflow

- Identify user and process problems.
- Define the Student / Course / Instructor / Enrollment data model.
- Configure tabs, Lightning App, record types and layouts.
- Configure validation rules and build flows.
- Configure the approval process.
- Create reports and the dashboard.
- Configure security.
- Test and document the evidence.

Milestone 1 - Data Model: Objects

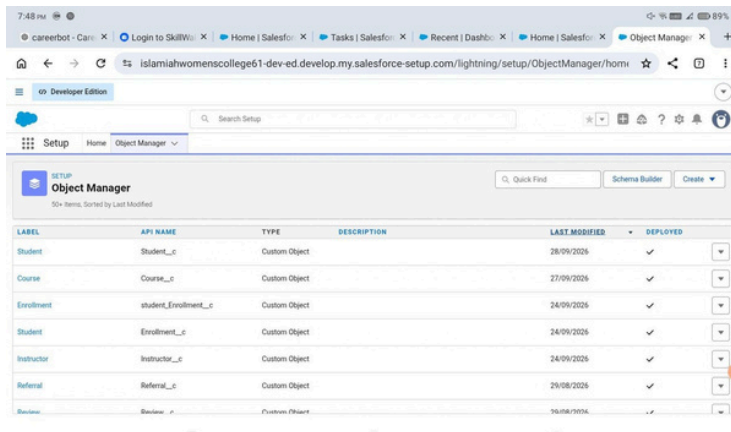
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Milestone 1 - Custom Objects

Four custom objects form the data layer: Student (personal and contact details, status), Course (category, duration, fee), Enrollment (transaction linking student, course and instructor) and Instructor (code, expertise, contact).

Development Process

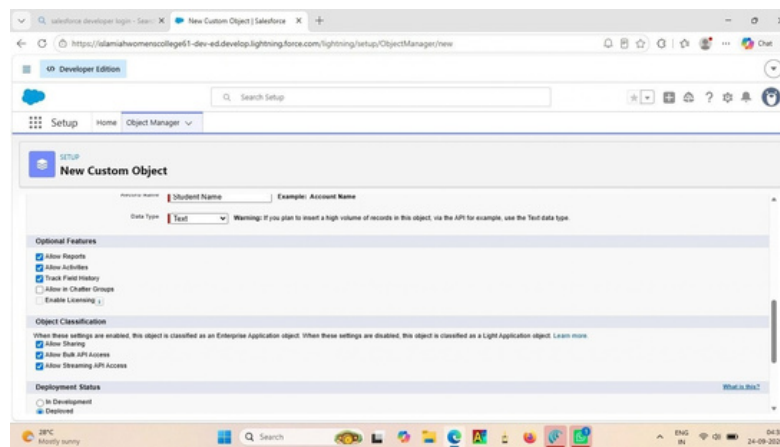
- OpenSetup>ObjectManager > Create > Custom Object.
- Setlabelandplurallabel (e.g. Student / Students) and the record name.
- Enablereporting,activities and search options; save and verify in Object Manager.



The screenshot shows the Salesforce Object Manager interface. A table lists the following custom objects:

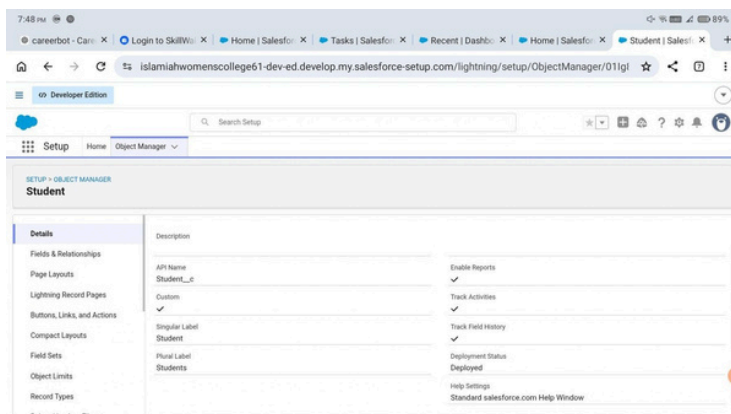
LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Student	Student__c	Custom Object		28/09/2026	✓
Course	Course__c	Custom Object		27/09/2026	✓
Enrollment	student_Enrollment__c	Custom Object		24/09/2026	✓
Student	Enrollment__c	Custom Object		24/09/2026	✓
Instructor	Instructor__c	Custom Object		24/09/2026	✓
Referral	Referral__c	Custom Object		29/08/2026	✓
Enrollment	Enrollment__c	Custom Object		16/08/2026	✓

Object Manager showing the SECMS custom objects



The screenshot shows the 'New Custom Object' form. The 'Object Name' is 'Student' and the 'API Name' is 'Student__c'. The 'Data Type' is 'Text'. The 'Optional Features' section includes checkboxes for 'Allow Reports', 'Allow Activities', 'Track Field History', 'Allow in Chatter Group', and 'Enable Learning'. The 'Object Classification' section includes checkboxes for 'Allow Sharing', 'Allow Bulk API Access', and 'Allow Streaming API Access'. The 'Deployment Status' is set to 'Deployed'.

Creating the Student custom object



The screenshot shows the 'Student' custom object configuration page. The 'Details' section includes fields for 'API Name', 'Singular Label', 'Plural Label', and 'Object Limits'. The 'Fields & Relationships' section includes a table for 'Fields' and a table for 'Relationships'. The 'Page Layouts' section includes a table for 'Page Layouts'. The 'Lightning Record Pages' section includes a table for 'Record Types'. The 'Buttons, Links, and Actions' section includes a table for 'Buttons, Links, and Actions'. The 'Compact Layouts' section includes a table for 'Compact Layouts'. The 'Field Sets' section includes a table for 'Field Sets'. The 'Object Limits' section includes a table for 'Object Limits'. The 'Record Types' section includes a table for 'Record Types'.

Student custom object

Milestone 1 - Data Model: Objects (continued)

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The remaining object screens are shown below:

The screenshot shows the Salesforce Setup interface for the 'Course' custom object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Course' and includes a 'Description' field, an 'API Name' field (Course__c), a 'Custom' checkbox (checked), a 'Singular Label' field (Course), a 'Plural Label' field (Course), and a 'Deployment Status' dropdown (Deployed). The right sidebar contains a 'Help Settings' section with a link to 'Standard salesforce.com Help Window'.

Course custom object

The screenshot shows the Salesforce Setup interface for the 'Enrollment' custom object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Enrollment' and includes a 'Description' field, an 'API Name' field (student_Enrollment__c), a 'Custom' checkbox (checked), a 'Singular Label' field (Enrollment), a 'Plural Label' field (Enrollments), and a 'Deployment Status' dropdown (Deployed). The right sidebar contains a 'Help Settings' section with a link to 'Standard salesforce.com Help Window'.

Enrollment custom object

The screenshot shows the Salesforce Setup interface for the 'Instructor' custom object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Instructor' and includes a 'Description' field, an 'API Name' field (Instructor__c), a 'Custom' checkbox (checked), a 'Singular Label' field (Instructor), a 'Plural Label' field (Instructors), and a 'Deployment Status' dropdown (Deployed). The right sidebar contains a 'Help Settings' section with a link to 'Standard salesforce.com Help Window'.

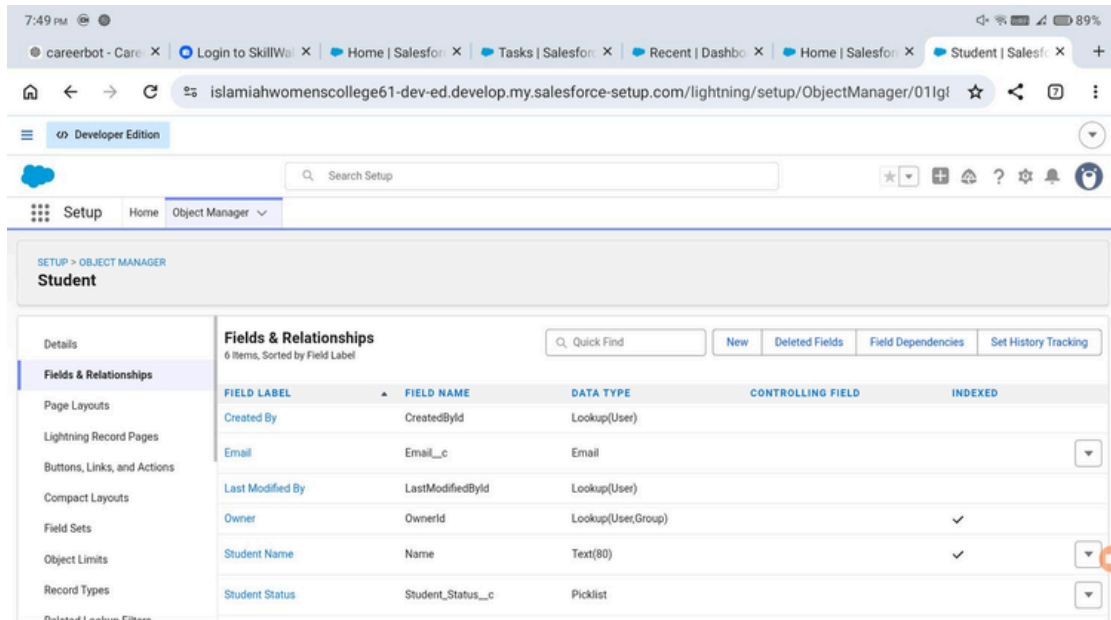
Instructor custom object

Milestone 2 - Fields

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Fields

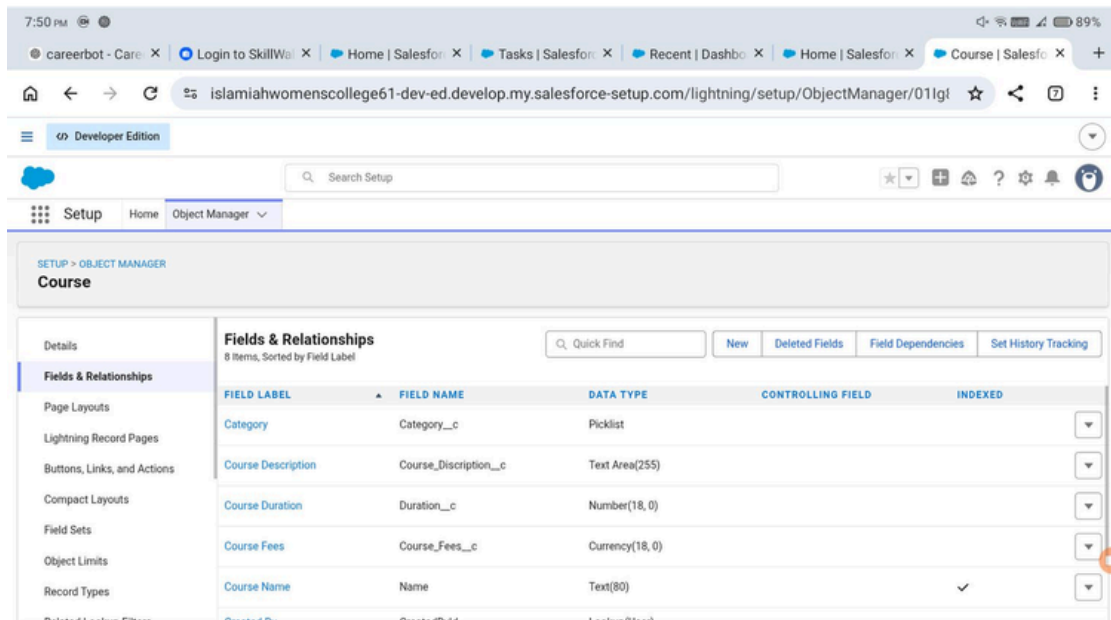
Student: Student Name, Email, Phone, Date of Birth, Student Status (picklist), Address. **Course:** Course Name, Category (picklist), Duration, Course Fee (currency), Description. **Instructor:** Instructor Name, Instructor Code, Expertise (picklist), Phone, Email. **Enrollment:** Student, Course and Instructor lookups, Enrollment Date, Enrollment Status (picklist), Fees Paid (checkbox), Total Amount (currency), Comments.



The screenshot shows the Salesforce Setup interface for the 'Student' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Fields & Relationships' and shows a table of 6 fields, sorted by Field Label. The table has columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are: Created By (Lookup(User)), Email (Email), Last Modified By (Lookup(User)), Owner (Lookup(User,Group)), Student Name (Text(80)), and Student Status (Picklist).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Student Name	Name	Text(80)		✓
Student Status	Student_Status__c	Picklist		

Student fields



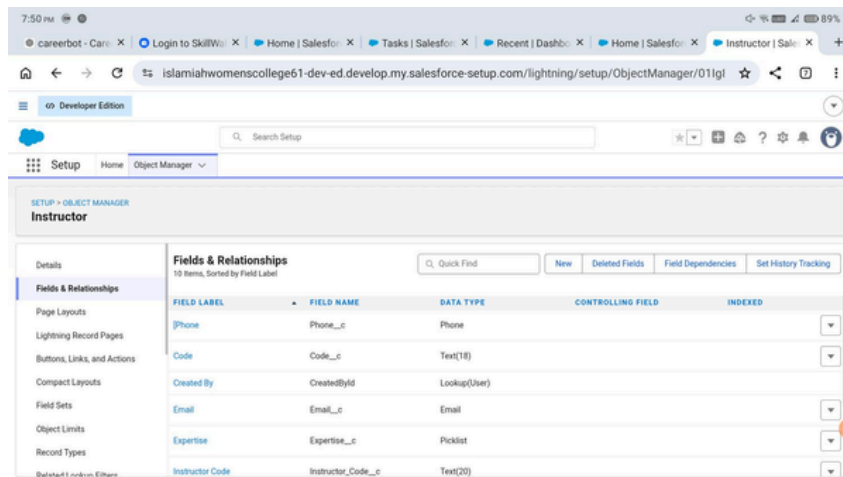
The screenshot shows the Salesforce Setup interface for the 'Course' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Fields & Relationships' and shows a table of 8 fields, sorted by Field Label. The table has columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are: Category (Picklist), Course Description (Text Area(255)), Course Duration (Number(18, 0)), Course Fees (Currency(18, 0)), Course Name (Text(80)), and Created By (Lookup(User)).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Category	Category__c	Picklist		
Course Description	Course_Description__c	Text Area(255)		
Course Duration	Duration__c	Number(18, 0)		
Course Fees	Course_Fees__c	Currency(18, 0)		
Course Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		

Course fields

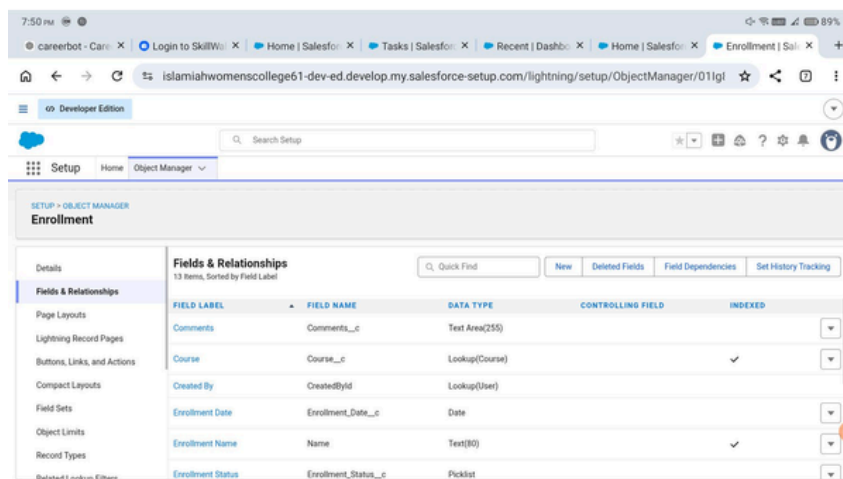
Milestone 2 - Fields (continued)

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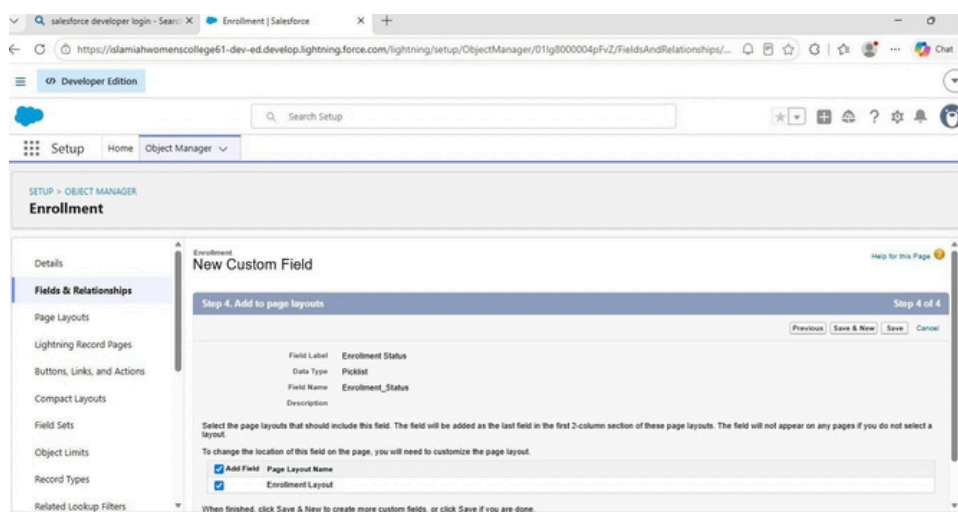
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Phone	Phone__c	Phone		
Code	Code__c	Text(18)		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Expertise	Expertise__c	Picklist		
Instructor Code	Instructor_Code__c	Text(20)		

Instructor fields



FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Comments	Comments__c	Text Area(255)		
Course	Course__c	Lookup(Course)		✓
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date__c	Date		
Enrollment Name	Name	Text(80)		✓
Enrollment Status	Enrollment_Status__c	Picklist		

Enrollment fields and lookup relationships



New Custom Field

Step 4. Add to page layouts

Field Label: Enrollment Status
Data Type: Picklist
Field Name: Enrollment_Status
Description:

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout.

☒ Add Field Page Layout Name
☒ Add Field Enrollment Layout

When finished, click Save & New to create more custom fields, or click Save if you are done.

Creating the Enrollment Status picklist field

Milestone 2 - Relationships

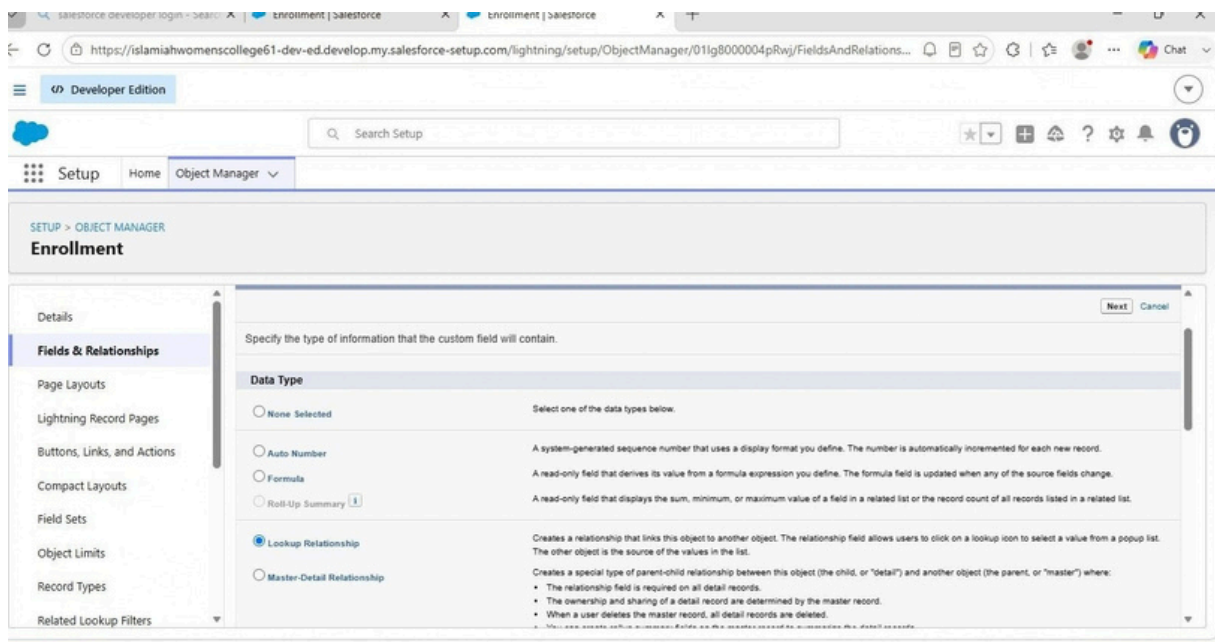
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Relationships

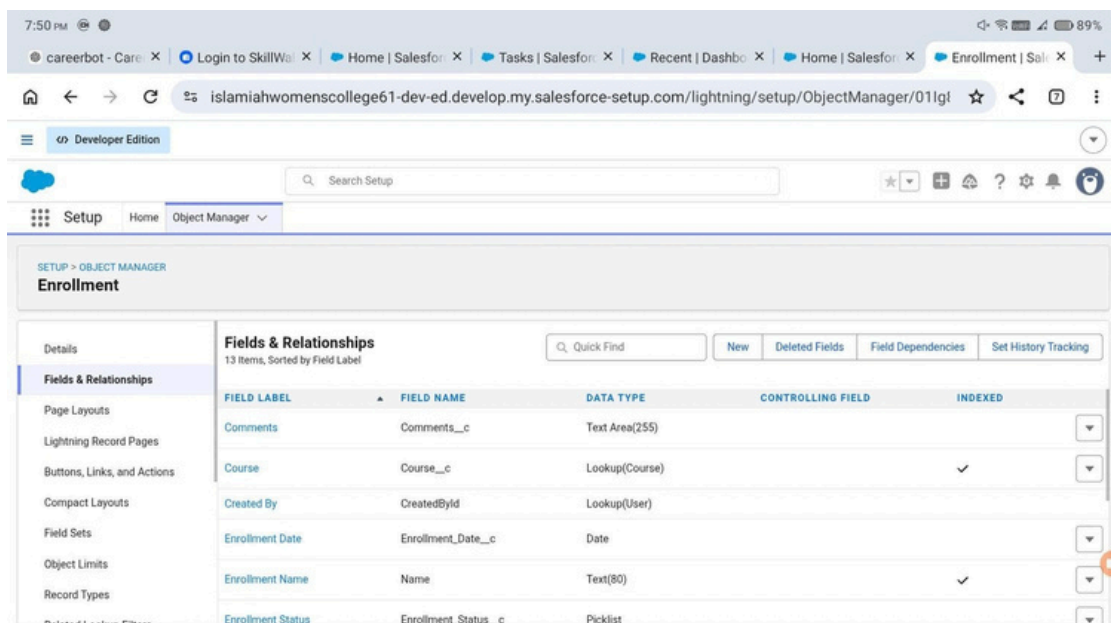
Lookup relationships from Enrollment to Student, Course and Instructor link each enrollment with its student, the selected course and the instructor. Sample relationship pattern: Student 1..* Enrollment *..1 Course; Enrollment *..1 Instructor.

Development Process

- Open Enrollment > Fields & Relationships > New.
- Choose Lookup Relationship and select the related object.
- Set the field label, visibility and page layout placement; save and verify.



Selecting Lookup Relationship data type on Enrollment



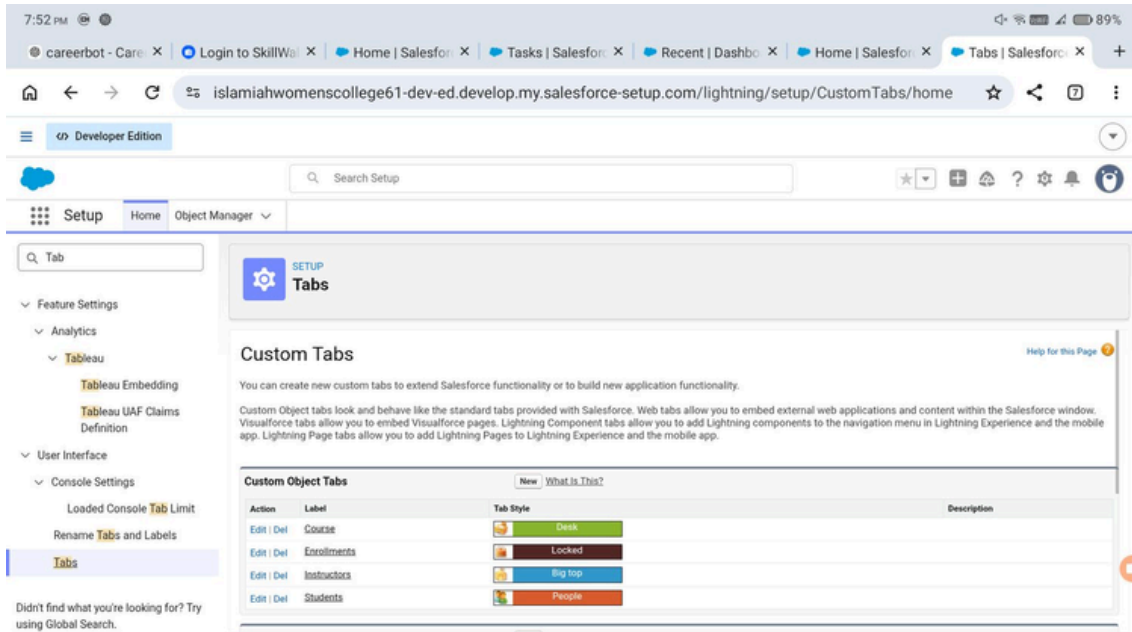
Enrollment fields and lookup relationships

Milestone 3 - Tabs and Lightning App

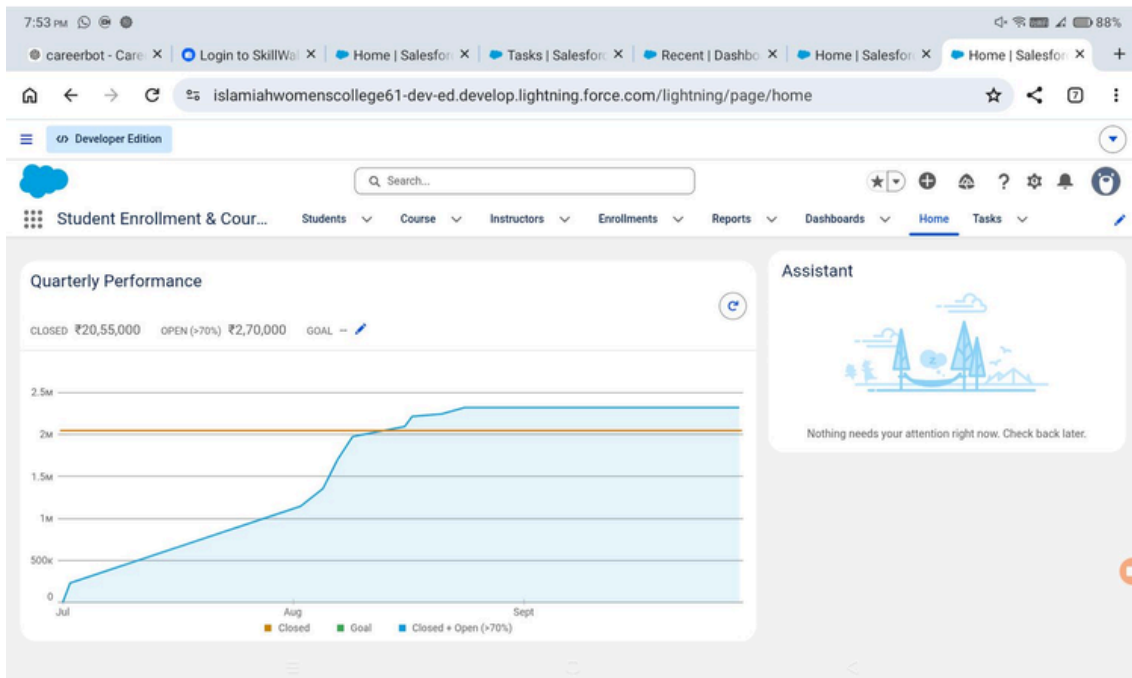
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Tabs and Lightning App

Custom tabs give direct navigation to Student, Course, Instructor and Enrollment. The Student Enrollment & Course Management Lightning App brings these tabs together with Home, Reports, Dashboards and Tasks in one interface.



Custom object tabs



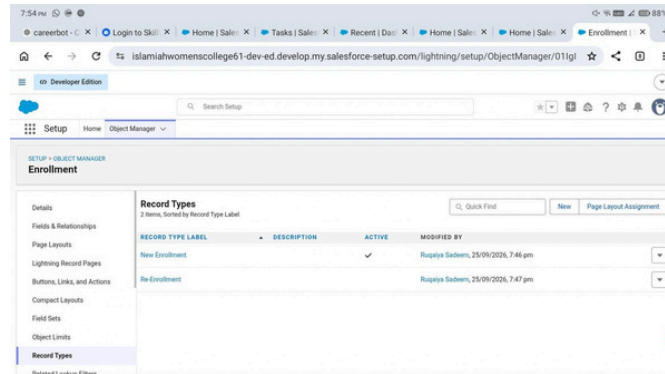
Student Enrollment & Course Management Lightning App

Milestone 4 - Record Types and Page Layouts

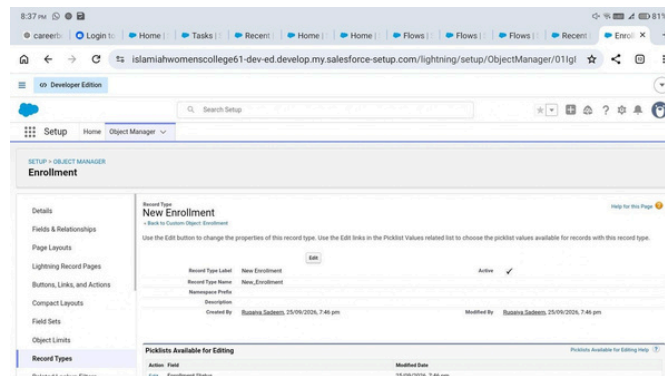
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Record Types and Page Layouts

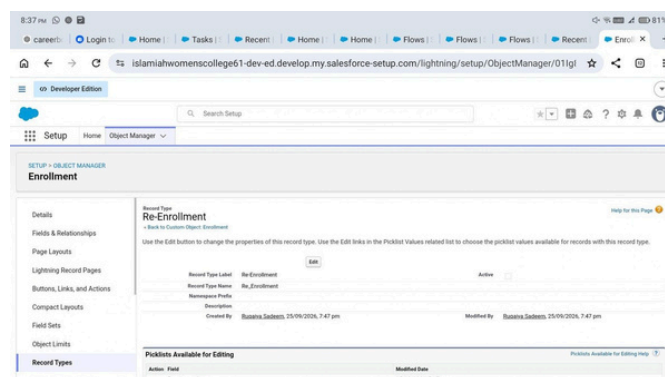
New Enrollment and Re-Enrollment record types separate first-time enrollment from returning-student processing. Each has its own page layout, and related lists show enrollment history on the Student, Course and Instructor pages.



Enrollment record types (New Enrollment, Re-Enrollment)



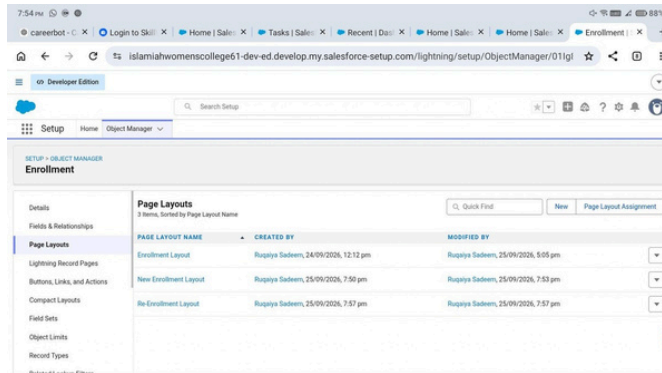
New Enrollment record type



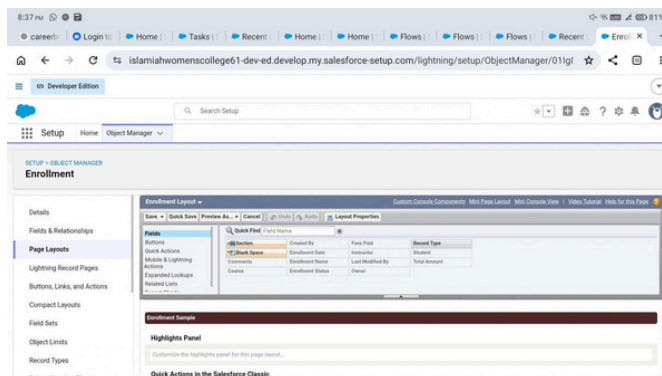
Re-Enrollment record type

Milestone 4 - Page Layouts (continued)

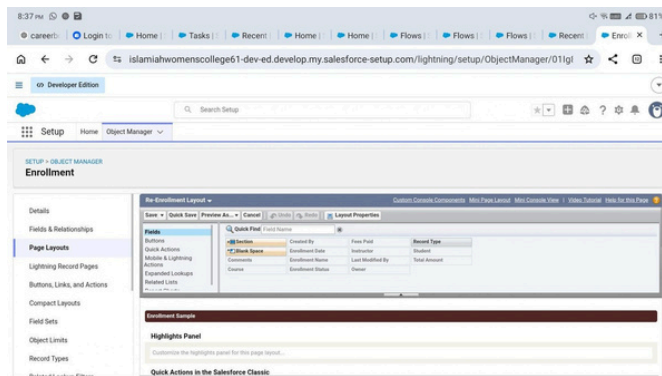
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Enrollment page layouts



Page layout editor (enrollment layout)



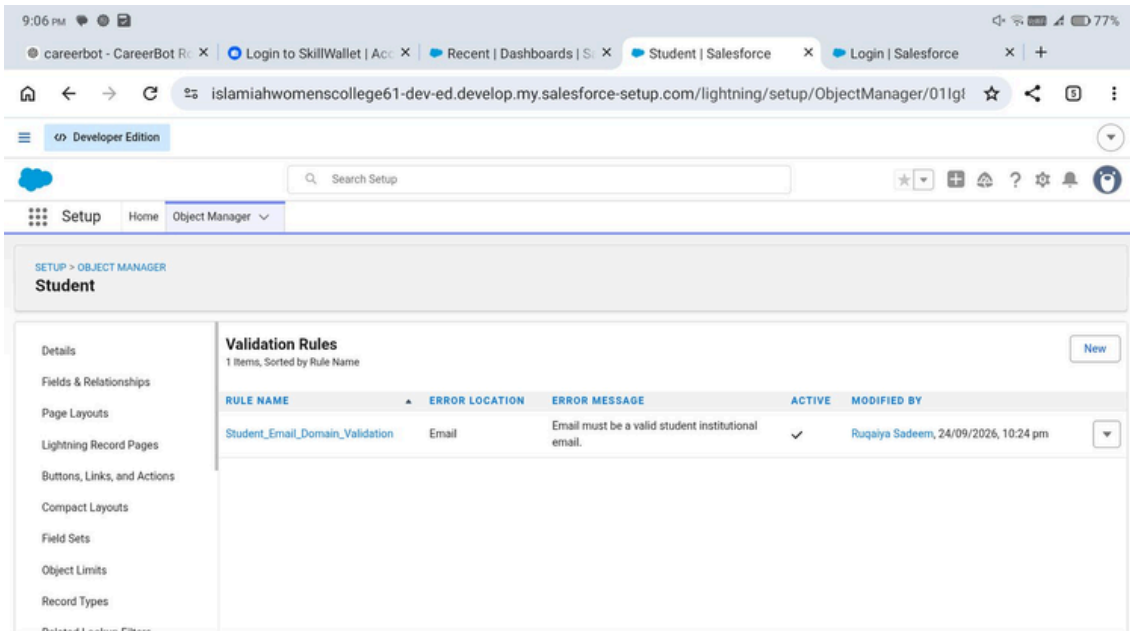
Page layout editor (Re-Enrollment layout)

Milestone 5 - Validation Rules

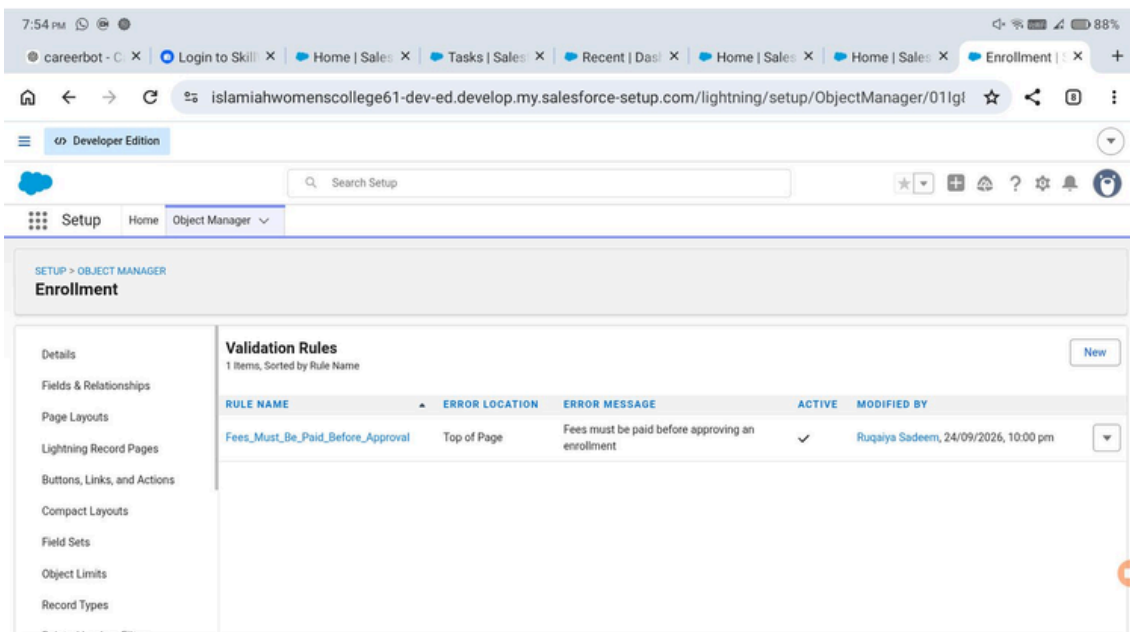
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Validation Rules

- **Student_Email_Domain_Validation**: checks that the student email meets the configured institutional-domain condition.
- **Fees_Must_Be_Paid_Before_Approval**: prevents an enrollment being marked Approved when Fees Paid is unchecked.



Student email validation rule



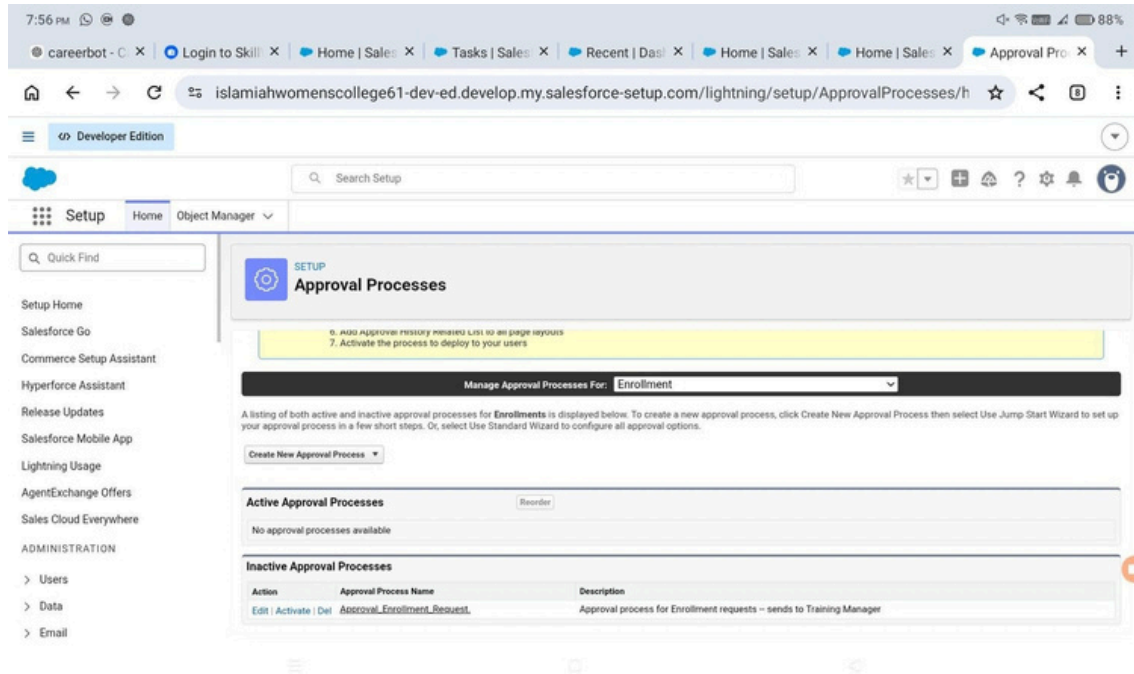
Enrollment fees-paid validation rule

Milestone 5 - Approval Process

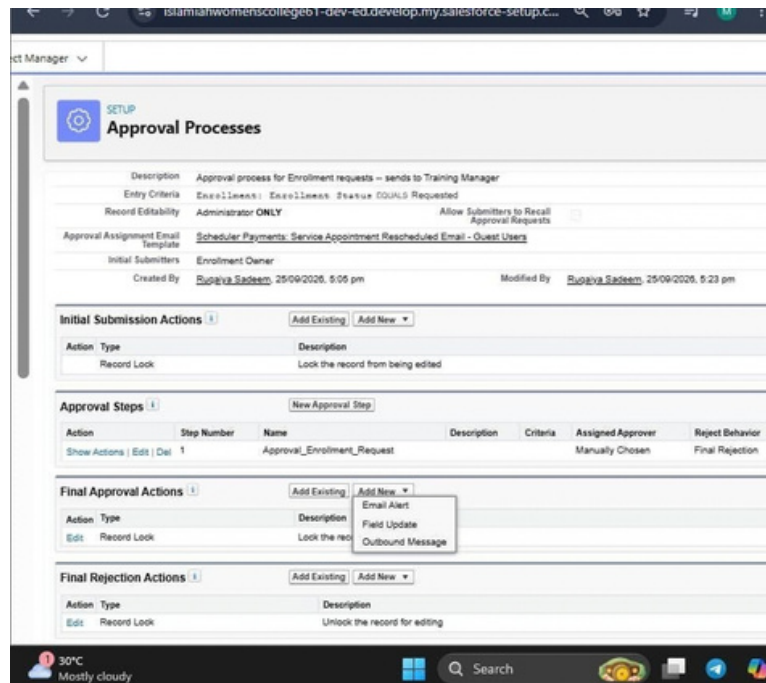
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Approval Process

Approval_Enrollment_Request routes enrollments with Enrollment Status = Requested to the designated approver. Final approval sets Enrollment_Status__c to Approved; final rejection sets the status to Rejected.



Approval processes for Enrollment



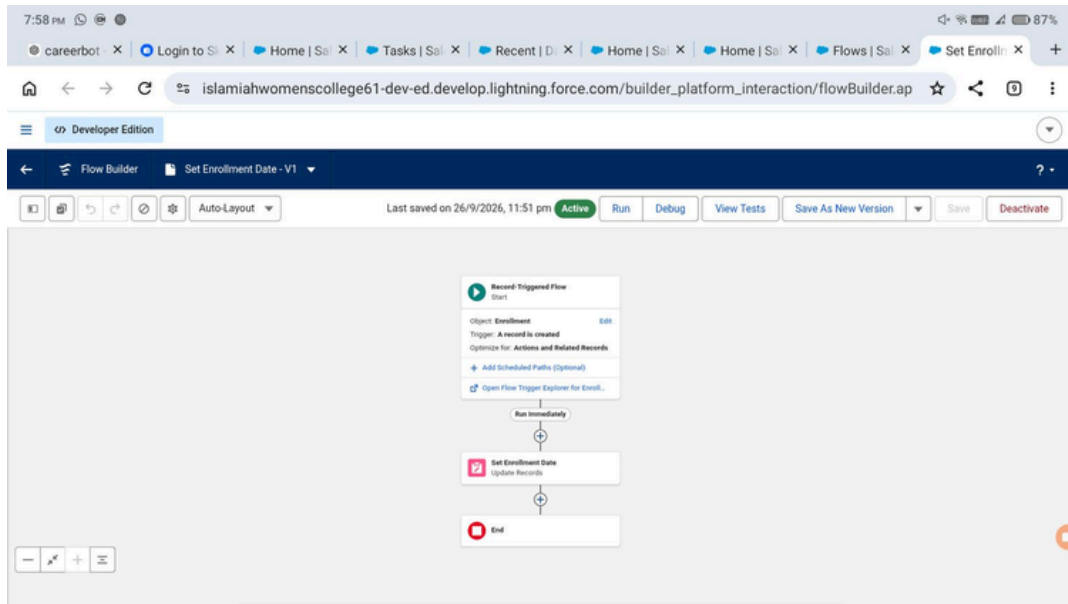
Approval process with final approval / rejection actions

Milestone 6 - Automation Flows

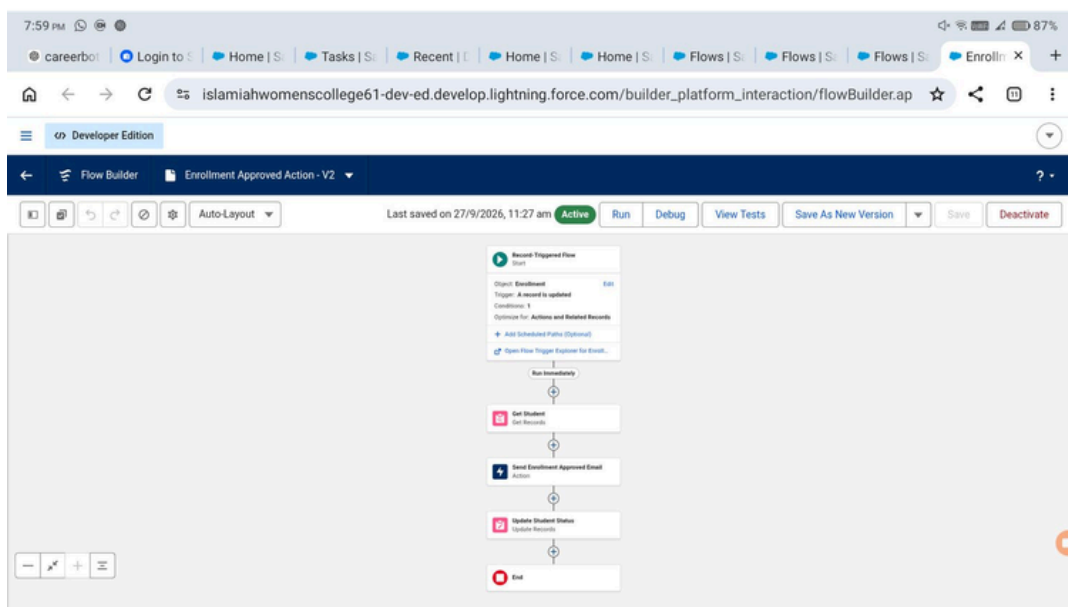
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Automation Flows

- SetEnrollmentDate: sets Enrollment_Date__c to the current date when an enrollment is created.
- EnrollmentApproved Action: notifies the student and activates the student after approval.
- AutoAssign_Instructor: assigns an instructor by course category (Technical, Language, Non-Technical).
- Create_Followup_Task: creates a task for the administrator when an enrollment stays in Requested status.



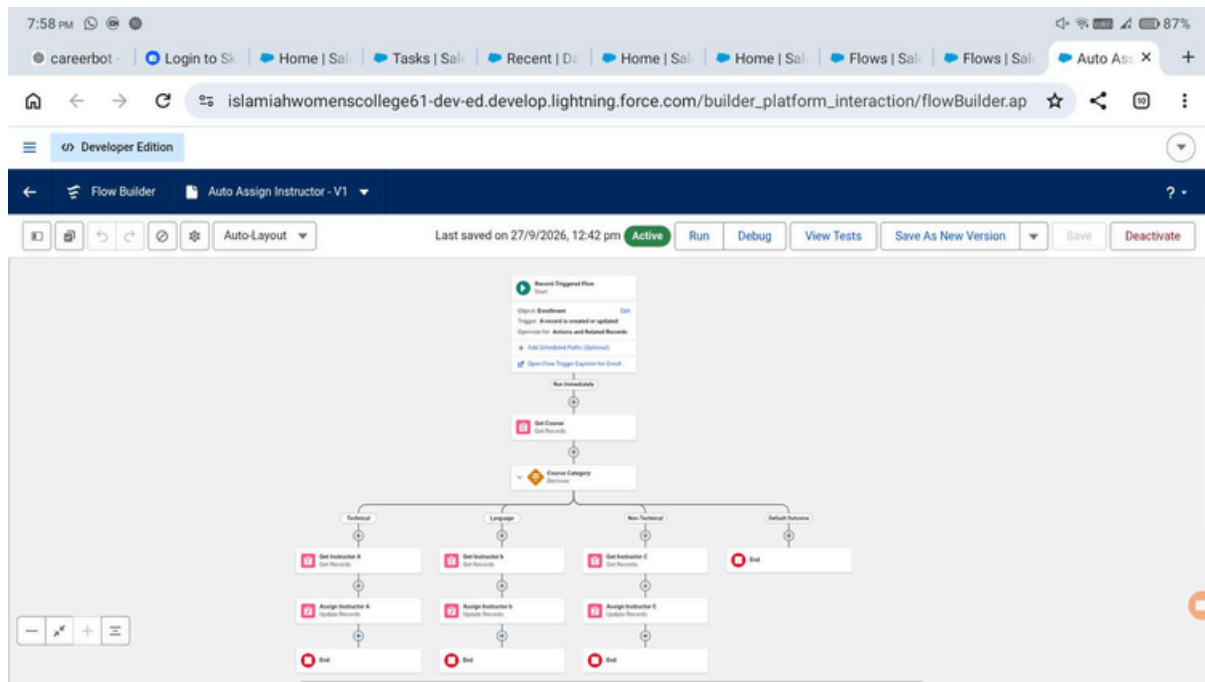
SetEnrollmentDate record-triggered flow



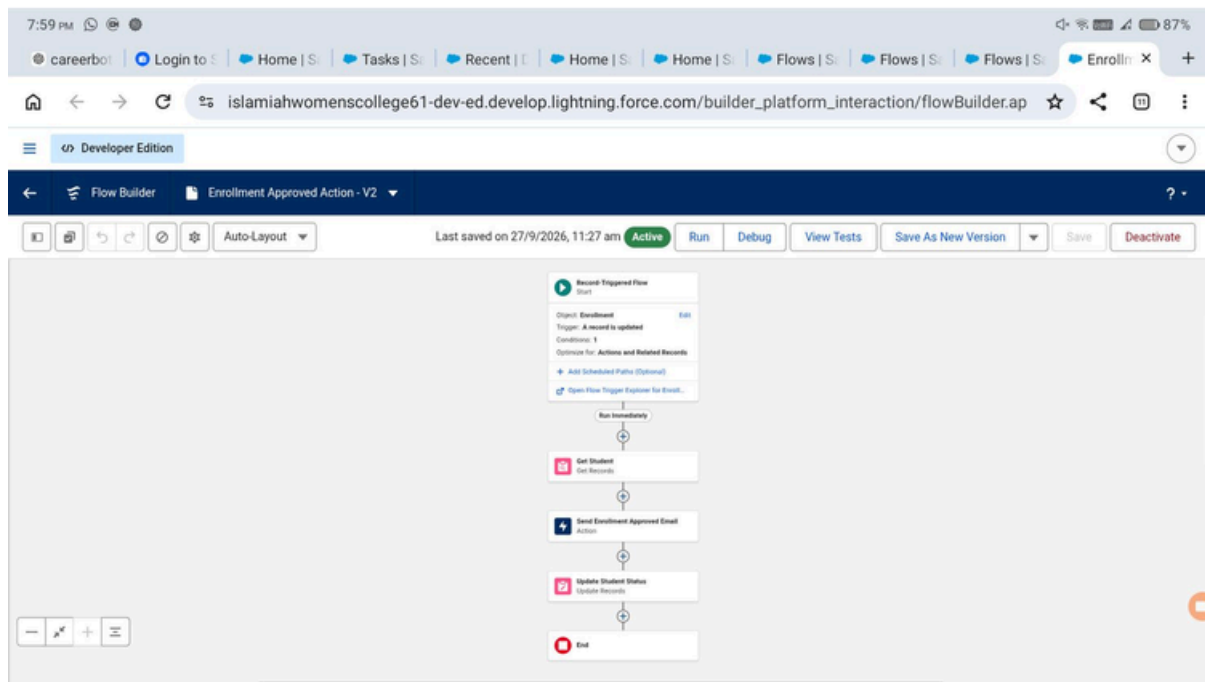
Enrollment Approved Action flow

Milestone 6 - Automation Flows (continued)

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AutoAssign_Instructor flow



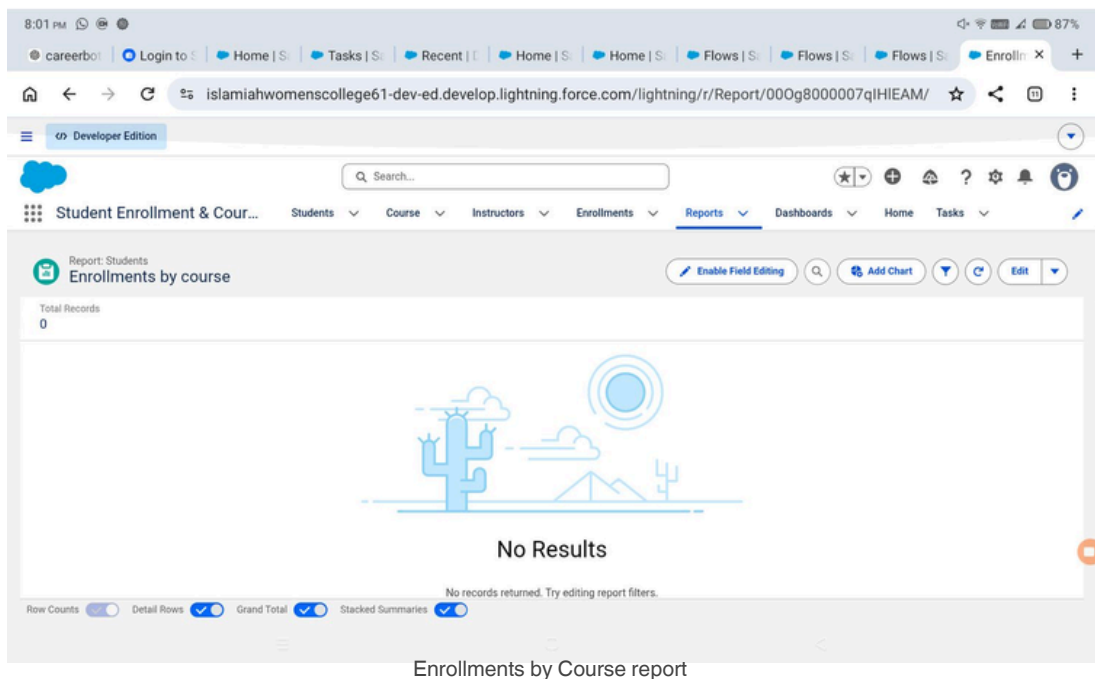
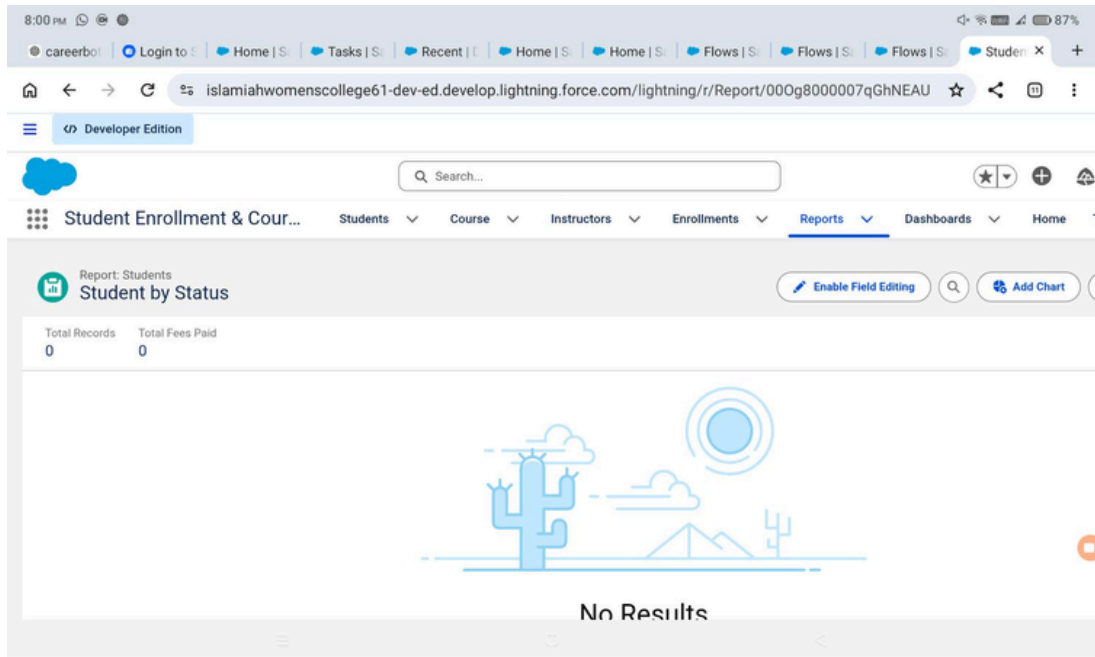
Flow builder screenshot supplied for the follow-up task flow

Milestone 7 - Reports

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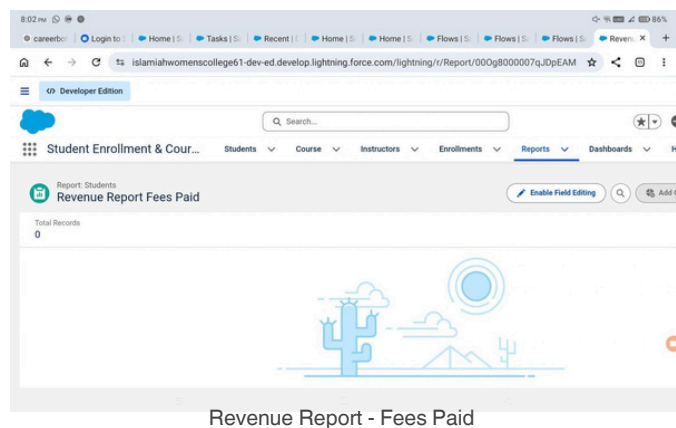
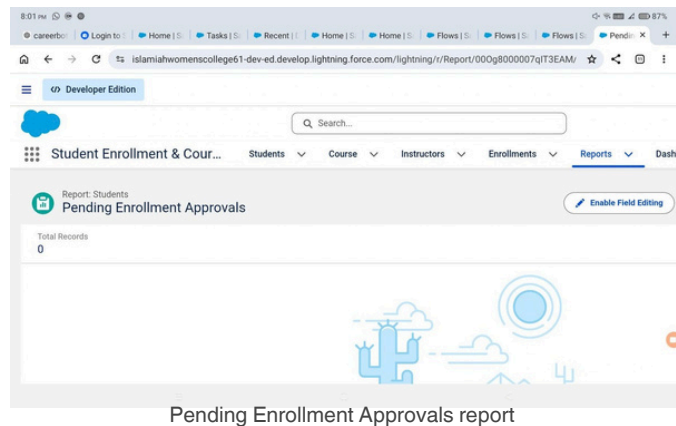
Reports

- Students by Status - students grouped by Student Status.
- Enrollments by Course - enrollment count per course.
- Pending Enrollment Approvals - enrollments in Requested status.
- Revenue Report - Fees Paid - paid enrollments summarized by course.



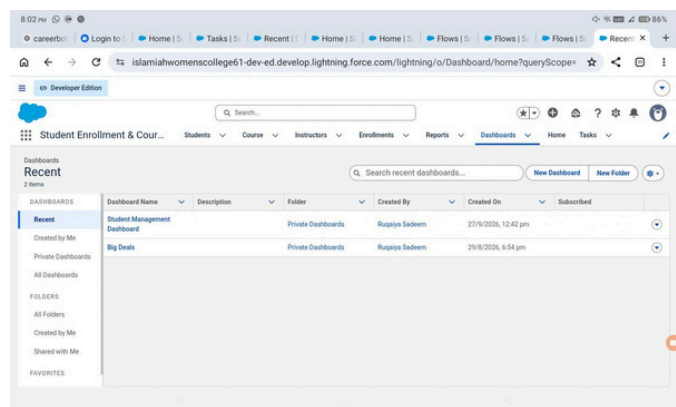
Milestone 7 - Reports and Dashboard (continued)

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Student Management Dashboard

Combines a donut chart (Students by Status), a bar chart (Enrollments by Course), a gauge (Revenue Report) and a table (Pending Enrollment Approvals).

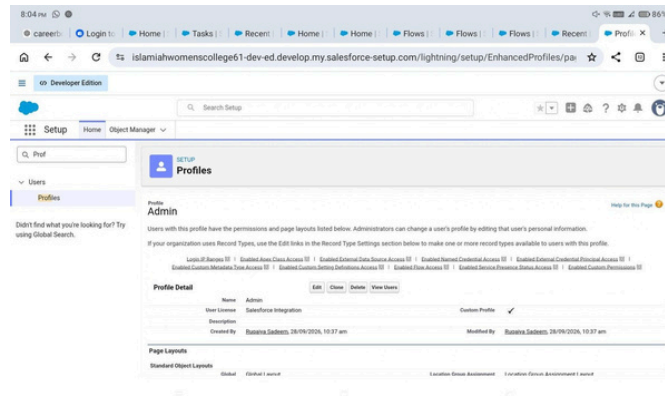


Milestone 8 - Security Configuration

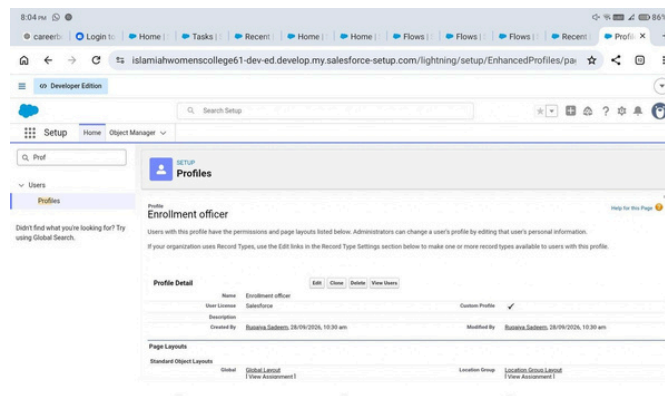
Group ID: SWTID-2026-4095 | Islamiah Women's Arts and Science College | Thiruvalluvar University

Security Configuration

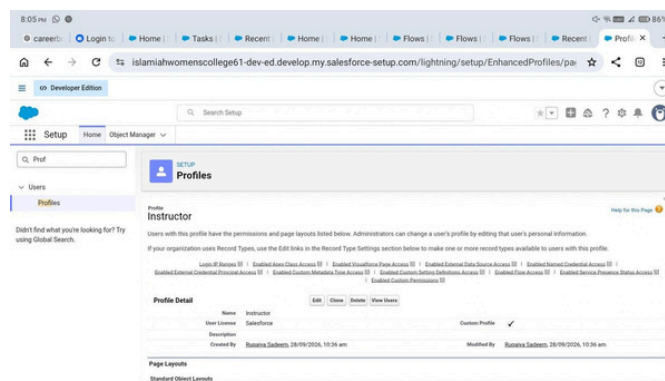
Profiles define baseline access for administrators, enrollment officers and instructors. Permission sets add access where required (for example custom object access and dashboard editing), and sharing settings / OWD control record visibility. The design supports least-privilege access.



Admin profile



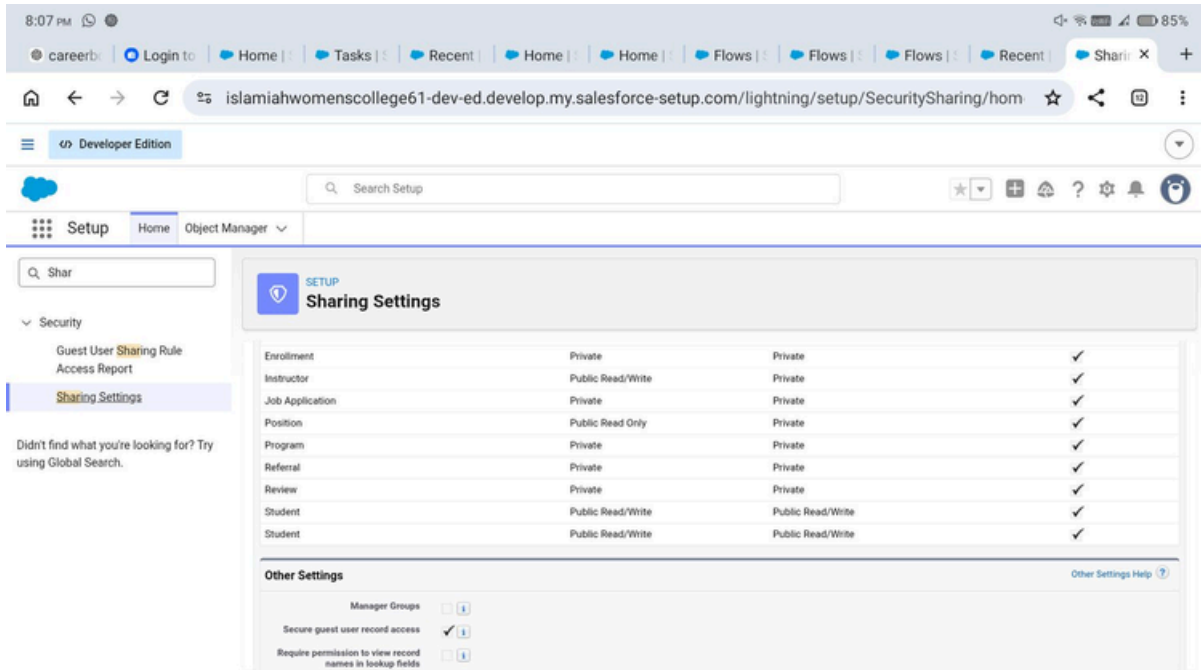
Enrollment Officer profile



Instructor profile

Milestone 8 - Sharing Settings

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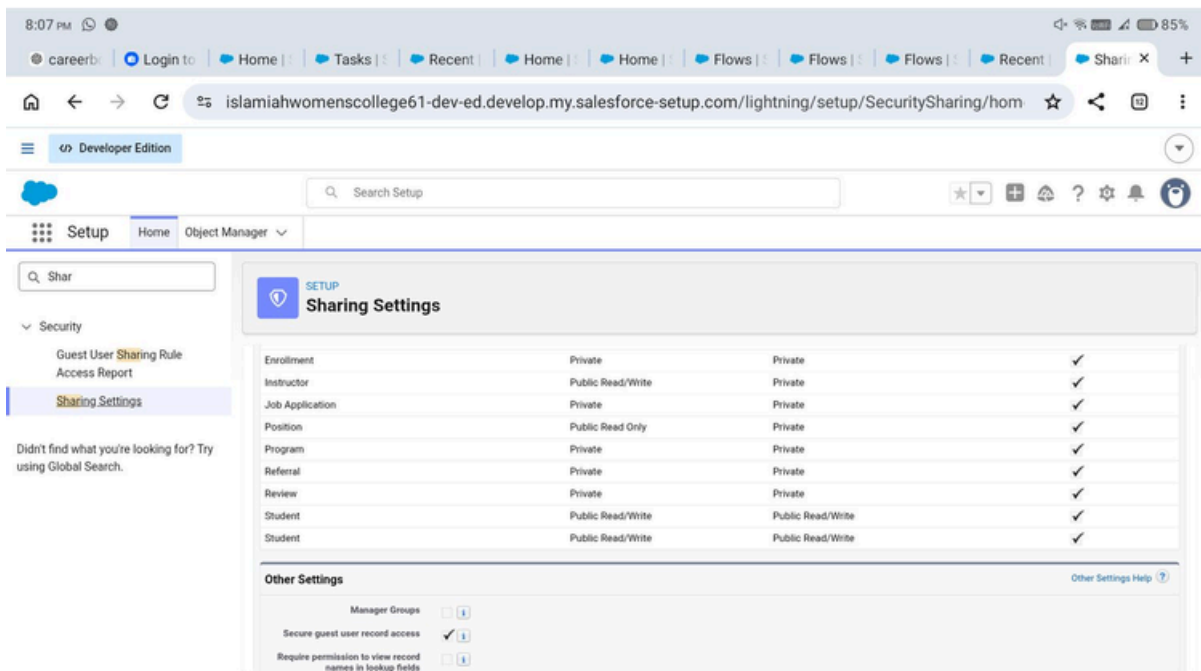
The screenshot shows the Salesforce Setup interface for the 'Sharing Settings' page. The left sidebar contains a search bar with 'Shar' and a list of security-related items, with 'Sharing Settings' highlighted. The main content area is titled 'Sharing Settings' and contains a table of sharing rules and 'Other Settings'.

Object	Private	Public Read/Write	Public Read Only
Enrollment	Private	Private	✓
Instructor	Public Read/Write	Private	✓
Job Application	Private	Private	✓
Position	Public Read Only	Private	✓
Program	Private	Private	✓
Referral	Private	Private	✓
Review	Private	Private	✓
Student	Public Read/Write	Public Read/Write	✓
Student	Public Read/Write	Public Read/Write	✓

Other Settings

- Manager Groups ☐ [?](#)
- Secure guest user record access ☒ [?](#)
- Require permission to view record names in lookup fields ☐ [?](#)

Sharing settings / organization-wide defaults



This is a duplicate of the screenshot above, showing the same Salesforce Setup - Sharing Settings page. It displays the table of sharing rules and the 'Other Settings' section.

Object	Private	Public Read/Write	Public Read Only
Enrollment	Private	Private	✓
Instructor	Public Read/Write	Private	✓
Job Application	Private	Private	✓
Position	Public Read Only	Private	✓
Program	Private	Private	✓
Referral	Private	Private	✓
Review	Private	Private	✓
Student	Public Read/Write	Public Read/Write	✓
Student	Public Read/Write	Public Read/Write	✓

Other Settings

- Manager Groups ☐ [?](#)
- Secure guest user record access ☒ [?](#)
- Require permission to view record names in lookup fields ☐ [?](#)

Sharing settings / organization-wide defaults (second view)

Testing Strategy and Test Cases

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Testing Strategy

- **Functional testing:** each feature performs its intended operation.
- **Validation testing:** invalid or incomplete data is prevented.
- **Automation testing:** flows and the approval process execute expected actions.
- **Reporting and dashboard testing:** reports and components show the configured data.
- **Security testing:** users receive only the intended access.

Representative Test Cases

- TC-01 to TC-10: create Student, Course, Instructor and Enrollment records; check relationships, record types and related lists.
- TC-11 Student email validation - validation message displayed (PASS).
- TC-12 Fees validation - Approved blocked when Fees Paid is false (PASS).
- TC-13 Enrollment date flow - date populated automatically (PASS).
- TC-14 Approval flow - notification / update actions execute (PASS).
- TC-15 Instructor assignment - configured instructor assigned (PASS).
- TC-16 Follow-up task - task created for Requested enrollment (PASS).
- TC-17 to TC-24: approval, rejection, reports and dashboard tests.
- SEC-01 to SEC-04: Admin, Enrollment Officer, Instructor and unauthorized-user access checks.

Full test tables are in the Project Report (chapter 7). Test statuses other than TC-11 to TC-16 are to be confirmed by the group after execution.

Advantages, Limitations, Conclusion and Future Scope

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Advantages

- Centralized academic data management.
- Automated enrollment workflows and structured approvals.
- Automatic instructor assignment by course category.
- Real-time reporting and dashboard visibility.
- Role-based access and cloud-based availability.

Limitations

- Dependence on the Salesforce platform.
- Initial configuration requires planning and administrative effort.
- Future integrations may need additional configuration or external services.

Conclusion

SECMS shows how Salesforce can support the academic enrollment lifecycle in one cloud platform. Centralized data reduces fragmentation, workflows reduce repetitive manual work, validation rules improve data quality, and reports and dashboards improve visibility.

Future Scope

- Phase1: payment gateway integration.
- Phase2: student self-service portal.
- Phase3: advanced analytics.
- Phase4: mobile access support.